

CutWise IMS – Inventory and Scrap Management Subsystem

Otto (SOG Manufacturing Corporation)

SPRINT BACKLOG
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PRODUCT BACKLOG – CutWise IMS: Inventory and Scrap Management Subsystem

(OTTO Shoes/SOG Manufacturing Corporation)

User Story ID	User Story	Acceptance Criteria	Priority (Sprint Backlog)	Requirements Reference	Revised Priority	Status
EPIC 1: AUTHENTICATION & ACCESS CONTROL						
US001	As an Admin, I want users to authenticate before accessing the system so that only authorized personnel can access inventory data.	1. User must enter valid username and password. 2. System validates credentials before login. 3. Invalid credentials display an error message. 4. Successful login redirects user to dashboard. 5. Session automatically expires after inactivity.	SPRINT 1	FR-001, FR-018		Completed
US002	As an Admin, I want role-based access control so that permissions are limited based on user roles.	1. System supports Admin, Co-Admin, Supervisor, and Inventory Clerk roles. 2. Unauthorized users cannot access restricted modules. 3. User permissions are configurable. 4. Access restrictions are enforced system-wide.	SPRINT 1	FR-018		Completed

US003	As an Admin, I want audit trail logging so that all user activities are recorded for monitoring purposes.	1. System records add, edit, delete, and configuration actions. 2. Audit logs include username, action, timestamp, and affected module. 3. Logs are viewable only by authorized users. 4. Audit trail cannot be manually modified.	SPRINT 2	FR-020	In Progress
EPIC 2: INVENTORY MANAGEMENT					
US004	As an Inventory Clerk, I want to add new raw materials so that inventory records remain updated.	1. User can input material name, type, size, quantity, unit cost, and supplier. 2. Required fields are validated before saving. 3. Duplicate entries are prevented. 4. Newly added material appears immediately in inventory list.	SPRINT 1	FR-001	In Progress
US005	As an Inventory Clerk, I want to update existing material information so that inventory records remain accurate.	1. Authorized users can edit material details. 2. Updated information reflects immediately after saving. 3. Changes are logged in audit trail. 4. System validates edited fields before update.	SPRINT 1	FR-002, FR-020	In Progress

US006	As a Supervisor, I want to view all inventory records so that I can monitor material availability.	1. Inventory table displays material name, type, quantity, size, and status. 2. Users can search materials. 3. Inventory list supports filtering. 4. Only authorized users can access inventory records.	SPRINT 1	FR-003	Completed
US007	As the system, I want inventory levels to update automatically so that stock records remain accurate after transactions.	1. Inventory deducts automatically after material usage. 2. Stock updates reflect in real-time. 3. Changes are recorded in audit logs. 4. Inventory balances cannot become negative.	SPRINT 2	FR-004	In Progress
US008	As a Supervisor, I want low stock alerts so that materials can be replenished before shortages occur.	1. System detects materials below threshold. 2. Email notification is sent to authorized users. 3. Alert is visible on dashboard. 4. Threshold values are configurable by Admin.	SPRINT 2	FR-005, FR-017	In Progress
EPIC 3: SCRAP MANAGEMENT					
US009	As the system, I want to record leftover leather materials from the cutting subsystem so that scrap inventory is tracked properly.	1. Scrap entries are automatically received from Sales & Cutting subsystem. 2. Scrap details include source batch and material type. 3. Scrap records are stored successfully. 4. Integration logs are recorded.	SPRINT 3	FR-006	In Progress

US010	As an Inventory Clerk, I want scrap materials converted into kilograms so that scrap inventory is standardized.	1. Scrap conversion records weight in kilograms. 2. System calculates equivalent scrap value. 3. Conversion history is stored. 4. Invalid weight entries are rejected.	SPRINT 3	FR-007	In Progress
US011	As a Supervisor, I want to view scrap inventory so that I can monitor reusable and sellable scraps.	1. Scrap inventory displays material type, weight, price/kg, and status. 2. Scrap inventory is searchable. 3. Only authorized users can access scrap records. 4. Scrap records update in real-time.	SPRINT 3	FR-008	In Progress
US012	As an Inventory Clerk, I want to record scrap sales so that scrap transactions are properly documented.	1. User can input buyer details and weight sold. 2. Scrap quantity deducts automatically after sale. 3. Sales history is recorded. 4. Sale transactions are logged in audit trail.	SPRINT 4	FR-009	In Progress
US013	As a Supervisor, I want profit analysis reports for scrap sales so that management can evaluate profitability.	1. System calculates total profit automatically. 2. Profit reports display trends and summaries. 3. Reports are accessible only to authorized users. 4. Reports can be exported.	SPRINT 4	FR-010, FR-015	In Progress
EPIC 4: SUPPLIER MANAGEMENT					

US014	As an Admin, I want to add supplier information so that supplier records are centralized.	1. User can input supplier name, contact details, and address. 2. Duplicate suppliers are prevented. 3. Supplier list updates immediately after saving. 4. Only authorized users can manage suppliers.	SPRINT 4	FR-011	Completed
US015	As a Supervisor, I want to view supplier information so that I can identify material sources easily.	1. Supplier records display contact details and materials supplied. 2. Users can search suppliers. 3. Supplier information is viewable only to authorized users. 4. Supplier records are updated in real-time.	SPRINT 4	FR-012	Completed
EPIC 5: REPORTS & ANALYTICS					
US016	As a Supervisor, I want to generate inventory reports so that I can monitor stock movement and inventory status.	1. Reports display current stock levels. 2. Reports include low-stock materials. 3. Reports are downloadable in PDF or Excel format. 4. Reports are filterable by date range.	SPRINT 5	FR-013	Not Yet Started
US017	As a Supervisor, I want to generate scrap reports so that scrap inventory and sales can be analyzed.	1. Reports display scrap quantity and sales history. 2. Reports include total scrap sold and remaining scrap. 3. Reports are downloadable. 4. Reports support date filtering.	SPRINT 5	FR-014	Not Yet Started

US018	As a Manager, I want profit analysis dashboards so that I can evaluate operational performance.	1. Dashboard displays scrap profit trends. 2. Charts and graphs are visible. 3. Reports update automatically. 4. Access is restricted to management users.	SPRINT 5	FR-015	Not Yet Started
EPIC 6: SETTINGS & CONFIGURATION					
US019	As an Admin, I want to configure general system settings so that operational preferences can be managed centrally.	1. Admin can update company name, currency, and measurement units. 2. Changes apply system-wide. 3. Configuration updates are logged. 4. Unauthorized users cannot modify settings.	SPRINT 6	FR-016	Not Yet Started
US020	As an Admin, I want to configure low stock thresholds so that notifications follow company standards.	1. Admin can define threshold values per material. 2. Threshold changes save successfully. 3. Updated thresholds affect future alerts. 4. Changes are recorded in audit logs.	SPRINT 6	FR-017	Not Yet Started
US021	As an Admin, I want to manage user accounts so that authorized personnel can access the system based on their roles.	1. Admin can create, update, deactivate, and reactivate users. 2. User roles determine accessible modules. 3. New users receive login credentials. 4. User management actions are audit logged.	SPRINT 6	FR-018	Not Yet Started

SPRINT BACKLOG 1

TIMELINE: MAY 11 - 18, 2025

Sprint Goal:Implement the core foundational functionalities of the CutWise IMS subsystem for OTTO Shoes Manufacturing, including secure authentication, role-based access control, inventory management, scrap management, supplier reference management, low stock monitoring, and audit trail logging.

User Story Number	User Story	Tasks	Acceptance Criteria / Task	Remarks (Story Points)
US001	As a User, I want to securely log in to the system so that only authorized personnel can access the subsystem.	[FE] Design Login UI [FE] Add validation and error handling [FE] Add show/hide password toggle [BE] Create POST /auth/login endpoint [BE] Implement JWT/session authentication [BE] Validate hashed passwords [BE] Implement failed login restrictions [BE] Validate user roles	• Login form contains Email/Username and Password fields • Invalid credentials display an error message • More than 5 failed login attempts temporarily lock the account • Users are redirected to role-based dashboards after successful login • JWT/session expires automatically after inactivity • Login page is responsive on desktop and mobile devices	5

US002	As an Admin, I want role-based access control so that users can only access authorized modules and functionalities.	[FE] Create role-based sidebar navigation [FE] Restrict unauthorized pages/modules [BE] Implement permission validation middleware [BE] Restrict sensitive modules to Admin and Supervisor	• Users only see modules accessible to their assigned role • Unauthorized users cannot access restricted pages • Inventory Clerk cannot access Settings, Profit Analysis, and Audit Trail modules • Only Admin can manage user accounts and configurations	3
US003	As an Inventory Clerk, I want to add new raw materials so that inventory records remain accurate and updated.	[FE] Build Add Material form UI [FE] Add validation for required fields [FE] Add success/error notifications [BE] Create POST /materials endpoint [BE] Prevent duplicate material entries [BE] Validate supplier reference [BE] Record activities in audit trail	• Form includes Material Name, Type, Size, Quantity, Unit Cost, and Supplier • Required fields are validated before submission • Duplicate material records are rejected • Materials cannot be saved without a valid supplier reference • Newly added material appears immediately in Inventory Management table • Add Material activity is logged with username and timestamp	5

US004	As an Inventory Clerk, I want to update material information so that inventory records remain accurate and updated.	[FE] Add Edit Material functionality [FE] Add Save and Cancel buttons [FE] Add validation for edited fields [BE] Create PATCH /materials/:id endpoint [BE] Record update actions in audit trail	• Authorized users can edit material details • Invalid or empty fields display validation errors • Updated material data reflects immediately after saving • Update actions are logged with username and timestamp	4
US005	As a Supervisor, I want to view inventory records so that I can monitor material availability and stock levels.	[FE] Build Inventory Management table UI [FE] Add search and filtering functionality [FE] Add stock status indicators [BE] Create GET /materials endpoint [BE] Implement pagination	• Inventory table displays material name, type, size, quantity, unit cost, supplier, and stock status • Users can search materials by name or type • Inventory can be filtered by material type and stock status • System displays In Stock, Low Stock, and Out of Stock labels • Inventory data updates in real time	4

US006	As a Supervisor, I want low stock alerts so that materials can be replenished before shortages occur.	[FE] Add low stock indicators [FE] Display low stock alerts on dashboard [BE] Implement configurable threshold logic [BE] Trigger low stock notifications	• Low stock materials are visually highlighted • Dashboard displays active low stock notifications • System detects materials below configured threshold • Notifications are sent to authorized users • Threshold values can be configured by Admin	3
US007	As the system, I want leftover leather materials from the Sales & Cutting subsystem to be recorded automatically so that scrap inventory is monitored properly.	[BE] Create scrap inventory database structure [BE] Create API integration endpoint [BE] Store source batch and material details [BE] Record scrap activities in audit trail [FE] Build Scrap Inventory UI	• Scrap data is received successfully from the Sales & Cutting subsystem • Scrap entries contain source batch, material type, and quantity details • Scrap inventory updates automatically after cutting transactions • Scrap records are displayed correctly in the Scrap Inventory module	5

US008	As an Inventory Clerk, I want unusable leather cut-offs converted into kilogram-based scrap inventory so that scrap inventory is standardized.	[FE] Build Convert Scrap form UI [BE] Implement scrap conversion logic [BE] Calculate scrap value automatically [BE] Store scrap conversion history [BE] Validate scrap weight entries	• Users can input scrap weight and details • System converts unusable leather into kilogram-based scrap inventory • Scrap value computation is accurate • Invalid weight entries are rejected • Converted scrap appears immediately in Scrap Inventory records	4
US009	As a Supervisor, I want to view scrap inventory so that I can monitor reusable and sellable scrap materials.	[FE] Build Scrap Inventory table UI [FE] Add search and filtering functionality [FE] Add scrap status indicators [BE] Create GET /scrap endpoint [BE] Restrict unauthorized access	• Scrap inventory displays material type, weight, value, and status • Users can search and filter scrap inventory records • Scrap inventory displays Available, Reserved, or Sold labels • Only authorized users can access scrap inventory	4

US010	As an Inventory Clerk, I want to add supplier information so that supplier records are centralized and updated.	[FE] Build Add Supplier form UI [FE] Add validation for supplier fields [BE] Create POST /suppliers endpoint [BE] Prevent duplicate supplier records [BE] Record supplier activities in audit trail	• Form includes Supplier Name, Contact Number, Email, and Address fields • Required supplier fields are validated before submission • Duplicate supplier records are rejected • Newly added supplier appears immediately in Supplier Reference module	3
US011	As an Admin, I want audit trail logging so that all user activities are traceable for monitoring and security purposes.	[FE] Build Audit Trail page UI [FE] Add filtering functionality [BE] Create audit_logs table [BE] Automatically record user activities [BE] Restrict access to Admin users only	• Audit Trail page displays activity logs correctly • Logs can be filtered by username, action type, and date • System logs add, update, delete, and configuration actions • Logs contain username, timestamp, and affected module • Unauthorized users cannot access audit logs	4
			TOTAL STORY PONTS	44

SPRINT BACKLOG 2

TIMELINE: MAY 20 - 26, 2025

Sprint Goal:Sprint Goal:Implement the secondary operational functionalities of the CutWise IMS subsystem including audit trail integration, automatic inventory updates, low stock notification system, scrap inventory processing,supplier management enhancements, and initial reporting functionalities for OTTO Shoes Manufacturing

User Story Number	User Story	Tasks	Acceptance Criteria / Task	Remarks (Story Points)
US003	As an Admin, I want audit trail logging so that all user activities are recorded for monitoring purposes.	[FE] Finalize Audit Trail page UI [FE] Add filtering and sorting functionality [BE] Automatically log add/edit/delete/configuration actions [BE] Restrict audit trail access to Admin only [BE] Store username, action type, timestamp, and affected module	• System automatically logs add, update, delete, and configuration actions • Logs display username, action, timestamp, and module affected • Only Admin users can access audit trail logs • Audit trail records cannot be manually edited or deleted • Audit logs are searchable and filterable by date and user	5

US007	As the system, I want inventory levels to update automatically so that stock records remain accurate after transactions.	[BE] Implement automatic stock deduction logic [BE] Prevent negative inventory balances [BE] Connect inventory updates to scrap transactions [FE] Display real-time inventory changes [BE] Record automatic stock updates in audit logs	• Inventory automatically updates after material usage or scrap conversion • Stock balances update in real time • Inventory balances cannot become negative • Automatic updates are recorded in audit trail logs • Updated inventory data reflects immediately on dashboard tables	5
US008	As a Supervisor, I want low stock alerts so that materials can be replenished before shortages occur.	[FE] Add low stock alert cards on dashboard [FE] Add visual indicators for low stock materials [BE] Implement configurable threshold settings [BE] Create low stock notification trigger system [BE] Integrate email notification support	• Materials below threshold are automatically detected • Dashboard displays active low stock alerts • Email notifications are sent to authorized users • Threshold values are configurable by Admin • Low stock materials are visually highlighted in inventory tables	4

US009	As the system, I want leftover leather materials from the Sales & Cutting subsystem to be recorded automatically so that scrap inventory is monitored properly.	[BE] Finalize Scrap Inventory database structure [BE] Create API integration endpoint for Sales & Cutting subsystem [BE] Store material source batch details [FE] Display scrap entries in Scrap Inventory module [BE] Record scrap generation activities in audit trail	• Scrap records are automatically received from Sales & Cutting subsystem • Scrap entries contain source batch and material details • Scrap inventory updates after cutting transactions • Scrap records are visible in Scrap Inventory module • Scrap generation activities are logged properly\	5
US010	As an Inventory Clerk, I want unusable leather cut-offs converted into kilogram-based scrap inventory so that scrap inventory is standardized.	[FE] Finalize Convert Scrap modal UI [BE] Implement scrap conversion computation logic [BE] Calculate scrap value automatically [BE] Store scrap conversion history [BE] Validate scrap weight inputs	• Users can input scrap weight successfully • System converts scrap into kilogram-based inventory • Scrap value calculations are accurate • Invalid weight values are rejected • Converted scrap appears immediately in Scrap Inventory records	4

US011	As a Supervisor, I want to view scrap inventory so that I can monitor reusable and sellable scrap materials.	[FE] Improve Scrap Inventory table UI [FE] Add search and filtering functionality [FE] Add scrap status indicators [BE] Create pagination support [BE] Restrict unauthorized access to scrap records	• Scrap inventory displays material type, weight, value, and status • Users can search and filter scrap inventory records • Scrap inventory shows Available, Reserved, and Sold labels • Unauthorized users cannot access scrap inventory • Scrap inventory updates in real time	4
US012	As an Inventory Clerk, I want to record scrap sales so that scrap transactions are properly documented.	[FE] Build Record Scrap Sale modal UI [FE] Add buyer information fields [BE] Create POST /scrap-sales endpoint [BE] Deduct sold scrap quantity automatically [BE] Record sales activities in audit trail	• Users can input buyer details and quantity sold • Scrap quantity automatically deducts after sale • Scrap sales history is recorded successfully • Sale transactions are logged in audit trail • Invalid quantity inputs are rejected	5

US014	As an Admin, I want to add supplier information so that supplier records are centralized and updated.	[FE] Improve Add Supplier form UI [FE] Add validation for supplier fields [BE] Prevent duplicate supplier records [BE] Record supplier activities in audit trail [FE] Display success/error notifications	• Supplier form validates required fields • Duplicate suppliers are rejected • Newly added suppliers appear immediately in Supplier Reference module • Supplier activities are logged in audit trail • Supplier data updates in real time	3
US015	As a Supervisor, I want to view supplier information so that I can identify material sources easily.	[FE] Improve Supplier Reference UI [FE] Add supplier search functionality [BE] Create GET /suppliers endpoint [BE] Restrict unauthorized access [FE] Add supplier material tags	• Supplier records display contact details and supplied materials • Users can search suppliers by name or material type • Supplier records update in real time • Only authorized users can access supplier information	3

US016	As a Supervisor, I want to generate inventory reports so that I can monitor stock movement and inventory status.	[FE] Build Inventory Reports page [FE] Add date filtering functionality [BE] Generate inventory summary reports [BE] Add PDF/Excel export support [FE] Add report charts and visualizations	• Inventory reports display stock levels and low stock materials • Reports support date filtering • Reports are downloadable in PDF and Excel formats • Inventory charts display correctly • Only authorized users can access reports	6
			TOTAL STORY PONTS	44

SPRINT BACKLOG 3

TIMELINE: MAY 12 - June 15, 2026

Sprint Goal:Finalize the advanced operational functionalities of the CutWise IMS subsystem by implementing reporting and analytics, profit analysis, system settings, user management, dashboard monitoring, audit trail enhancement, and full frontend-backend integration to support initial reporting functionalities for OTTO Shoes Manufacturing

User Story Number	User Story	Tasks	Acceptance Criteria / Task	Remarks (Story Points)
US013	As a Supervisor, I want to monitor scrap profit analysis so that I can evaluate scrap sales performance and operational profitability.	[FE] Build Profit Analysis dashboard section [FE] Add scrap profit charts and analytics visualization [FE] Add date filtering functionality [BE] Calculate total scrap profit automatically [BE] Generate scrap sales summaries and trends [BE] Restrict access to authorized management users only [BE] Record profit analysis access in audit logs	• System calculates scrap profit accurately based on scrap sales transactions • Dashboard displays profit summaries, trends, and analytics charts • Profit analytics update automatically after scrap sales transactions • Reports support date filtering functionality • Unauthorized users cannot access the Profit Analysis module • Profit analytics display correctly in real time	5
US014	As a Supervisor, I want to generate scrap reports so that I can monitor scrap inventory, scrap sales, and scrap movement records efficiently.	[FE] Build Scrap Reports page [FE] Add report preview modal [FE] Add report filtering by date and scrap status [BE] Generate scrap inventory summaries [BE] Generate scrap sales summaries [BE] Add PDF and Excel export support [BE] Restrict unauthorized access to reports	• Scrap reports display scrap quantity, scrap value, sales records, and inventory summaries • Reports support filtering by date and scrap status • Reports are downloadable in PDF and Excel formats • Report previews display correctly before export • Reports update automatically based on current records • Only authorized users can access scrap reports	5
US015	As a Supervisor, I want to generate inventory reports so that I can monitor inventory status, stock movement, and material availability.	[FE] Build Inventory Reports page [FE] Add inventory summary cards and charts [FE] Add filtering by material type and stock status [BE] Generate inventory summaries [BE] Add PDF and Excel export functionality [BE] Integrate report data with inventory records	• Inventory reports display material quantity, stock status, and low stock materials • Reports support filtering by material type and stock status • Reports are downloadable in PDF and Excel formats • Inventory charts display correctly • Reports reflect real-time inventory data updates	5
US016	As an Admin, I want to configure system settings so that operational configurations and company preferences can be managed centrally.	[FE] Build System Settings module [FE] Add company information configuration form [FE] Add currency and measurement unit settings [FE] Add save confirmation modal [BE] Save configuration settings in database [BE] Restrict access to Admin users only [BE] Record configuration changes in audit trail	• Admin users can configure company name, currency format, and measurement units • Settings changes save successfully • Configuration changes are recorded in audit logs • Unauthorized users cannot access the Settings module • Updated settings reflect system-wide immediately after saving	4
US017	As an Admin, I want to configure low stock threshold values so that inventory alerts follow operational requirements.	[FE] Build threshold configuration form [FE] Add threshold validation messages [BE] Save low stock threshold values [BE] Integrate threshold settings with notification logic [BE] Record threshold changes in audit logs	• Admin users can configure low stock threshold values successfully • Threshold changes reflect immediately in the Low Stock Alert module • Low stock notifications follow updated threshold values • Invalid threshold inputs are rejected • Threshold configuration activities are recorded in audit logs	4
US018	As an Admin, I want to manage user accounts so that user access and permissions remain controlled and updated.	[FE] Build User Management module [FE] Add Add/Edit User modal [FE] Add role assignment dropdown [FE] Add user activation/deactivation functionality [BE] Create user management endpoints [BE] Implement role update functionality [BE] Restrict access to Admin users only [BE] Record user management activities in audit trail	• Admin users can add, edit, disable, and manage user accounts • User roles can be assigned and updated successfully • Inactive users cannot access the subsystem • Unauthorized users cannot access User Management module • User management activities are recorded in audit trail logs	6
US019	As a Supervisor, I want dashboard analytics so that I can quickly monitor inventory and scrap operations in real time.	[FE] Build Dashboard analytics cards [FE] Add inventory summary charts [FE] Add low stock monitoring cards [FE] Add scrap inventory analytics section [FE] Add supplier summary widgets [BE] Create dashboard analytics endpoints [BE] Aggregate inventory and scrap statistics	• Dashboard displays inventory summaries and low stock materials • Dashboard displays scrap inventory analytics and supplier summaries • Charts and analytics update automatically in real time • Dashboard loads successfully without performance issues • Dashboard widgets reflect accurate operational data	5

US020	As the system, I want frontend and backend integration finalized so that subsystem functionalities operate correctly end-to-end.	[BE] Finalize API integration for Inventory, Scrap, Supplier, Reports, and Audit modules [BE] Complete database migrations [FE] Connect frontend forms to backend APIs [FE] Validate real-time data synchronization [QA] Perform integration testing and validation [QA] Fix integration and synchronization issues	• Frontend modules successfully communicate with backend APIs • Inventory, Scrap, Supplier, Reports, and Audit modules synchronize correctly • Database transactions save and retrieve data successfully • System passes integration testing without critical issues • Real-time updates display correctly across modules	7
US021	As a User, I want improved responsiveness and usability so that the subsystem is easier to navigate and use across supported web browsers and screen resolutions	[FE] Optimize dashboard responsiveness [FE] Improve modal and table layouts [FE] Improve loading indicators and notifications [FE] Optimize mobile responsiveness [FE] Improve user interface consistency across modules	• Dashboard is responsive on desktop. • Tables and modals display correctly on different screen sizes • Loading states and notifications display correctly • UI consistency is maintained across all modules • User navigation experience is improved	4

TOTAL STORY POINTS : 45

INITIAL REQUIREMENTS TRACEABILITY MATRIX (RTM)				
CutWise IMS: Inventory and Scrap Management Subsystem				
OTTO Shoes Manufacturing				
Requirement ID	Business Requirement	Functional Requirement Reference	Initial Test Case Reference	Status
BR-001	The system must authenticate users before allowing access to the subsystem to ensure that only authorized personnel can use the system.	FR-018 User Authentication	TC-001 User Login Validation	Initial
BR-002	The system must implement role-based access control to restrict user permissions based on assigned roles such as Admin, Co-Admin, Supervisor, and Inventory Clerk.	FR-019 Role-Based Access Control	TC-002 Role Permission Validation	Initial
BR-003	The system must allow authorized users to add new raw materials with complete details such as material name, type, size, quantity, unit cost, and supplier information.	FR-001 Add Material	TC-003 Add Material Validation	Initial
BR-004	The system must allow authorized users to update existing material information to maintain accurate and updated inventory records.	FR-002 Update Material	TC-004 Update Material Validation	Initial
BR-005	The system must display inventory records including material name, type, size, quantity, stock status, and supplier information for monitoring purposes.	FR-003 View Inventory	TC-005 Inventory Viewing Validation	Initial
BR-006	The system must automatically update inventory levels whenever materials are added, modified, converted into scrap, or used in transactions.	FR-004 Real-Time Stock Update	TC-006 Inventory Auto-Update Validation	Initial
BR-007	The system must notify authorized users whenever material stock falls below the configured threshold level.	FR-005 Low Stock Alert	TC-007 Low Stock Notification Validation	Initial
BR-008	The system must record leftover leather materials received from the Sales & Cutting subsystem to support scrap inventory monitoring.	FR-006 Record Scrap	TC-008 Scrap Recording Validation	Initial
BR-009	The system must convert unusable leather cut-offs into kilogram-based scrap inventory for standardized scrap monitoring.	FR-007 Convert Scrap to KG	TC-009 Scrap Conversion Validation	Initial
BR-010	The system must display scrap inventory records including material type, weight, value, and scrap status.	FR-008 View Scrap Inventory	TC-010 Scrap Inventory Viewing Validation	Initial
BR-011	The system must allow authorized users to record scrap sales transactions and automatically update scrap inventory balances.	FR-009 Record Scrap Sale	TC-011 Scrap Sale Recording Validation	Initial
BR-012	The system must calculate and display scrap profit based on recorded scrap sales transactions.	FR-010 Track Scrap Profit	TC-012 Scrap Profit Calculation Validation	Initial
BR-013	The system must allow authorized users to add supplier records including supplier name, contact details, and supplied materials.	FR-011 Add Supplier	TC-013 Supplier Creation Validation	Initial
BR-014	The system must display supplier records and related material information for supplier monitoring and reference purposes.	FR-012 View Supplier	TC-014 Supplier Viewing Validation	Initial
BR-015	The system must generate downloadable inventory reports for monitoring inventory status, stock movement, and material availability.	FR-013 Generate Inventory Report	TC-015 Inventory Report Generation Validation	Initial
BR-016	The system must generate downloadable scrap reports including scrap quantity, scrap sales, and scrap inventory summaries.	FR-014 Generate Scrap Report	TC-016 Scrap Report Generation Validation	Initial

BR-017	The system must display scrap profit analysis and trends to support monitoring and decision-making.	FR-015 Profit Analysis	TC-017 Profit Analysis Validation	Initial
BR-018	The system must allow Admin users to configure system settings such as company information, currency format, and measurement units.	FR-016 Configure System Settings	TC-018 System Configuration Validation	Initial
BR-019	The system must allow Admin users to configure low stock threshold values for inventory monitoring.	FR-017 Set Low Stock Threshold	TC-019 Low Stock Threshold Configuration Validation	Initial
BR-020	The system must automatically record user activities such as add, update, delete, login, and configuration changes for monitoring and accountability purposes.	FR-020 Audit Trail Logging	TC-020 Audit Trail Logging Validation	Initial